WORKED SOLUTION

STEP I 2000, Question 5

Pure • Euclidean geometry

STEP I 2000, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q5

The key idea

Everything happens in the inclined plane, so set up two perpendicular unit vectors *in that plane*: one pointing straight up the line of greatest slope, the other horizontal. Arthur's and Bertha's positions are then simple combinations of these, and the sum-to-product formulae tidy up the difference at once. For the angle with the vertical, we only need to know how each of the two in-plane directions sits relative to the vertical, and that is where θ enters.

Solution

Setting up

Let $\mathbf{e}$ be the unit vector pointing up the steepest line through O, and let $\mathbf{f}$ be the unit vector in the plane, perpendicular to $\mathbf{e}$, on the side where both walkers are. Since $\mathbf{f}$ is perpendicular to the line of greatest slope, it is horizontal. Let $\mathbf{k}$ be the unit vector vertically upwards. Then

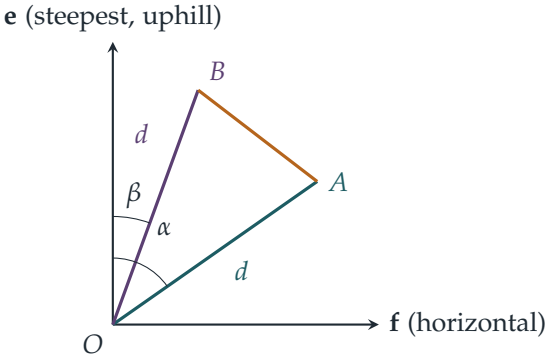
$$\mathbf{e}\cdot\mathbf{k} = \sin\theta, \qquad \mathbf{f}\cdot\mathbf{k} = 0, \qquad \mathbf{e}\cdot\mathbf{f} = 0.$$

(The first holds because $\mathbf{e}$ makes angle θ with the horizontal, hence angle $\frac{\pi}{2} - \theta$ with $\mathbf{k}$.)

Both walk at the same speed, so when Arthur has walked a distance d, so has Bertha. Their positions relative to O are

$$\mathbf{a} = d(\cos\alpha\,\mathbf{e} + \sin\alpha\,\mathbf{f}), \qquad \mathbf{b} = d(\cos\beta\,\mathbf{e} + \sin\beta\,\mathbf{f}).$$

Walking uphill on the same side of the steepest line means $0 \le \alpha, \beta < \frac{\pi}{2}$.



The figure is drawn in the inclined plane itself, looking straight at it.

The distance between them

Using the sum-to-product formulae,

$$\begin{aligned}\mathbf{a}-\mathbf{b} &= d[(\cos\alpha-\cos\beta)\,\mathbf{e}+(\sin\alpha-\sin\beta)\,\mathbf{f}]\\ &= d[-2\sin\tfrac{1}{2}(\alpha+\beta)\sin\tfrac{1}{2}(\alpha-\beta)\,\mathbf{e}+2\cos\tfrac{1}{2}(\alpha+\beta)\sin\tfrac{1}{2}(\alpha-\beta)\,\mathbf{f}]\\ &= 2d\sin\tfrac{1}{2}(\alpha-\beta)\,[-\sin\tfrac{1}{2}(\alpha+\beta)\,\mathbf{e}+\cos\tfrac{1}{2}(\alpha+\beta)\,\mathbf{f}].\end{aligned}$$

Write $s=\frac{1}{2}(\alpha+\beta)$. The vector $\mathbf{u}=-\sin s\,\mathbf{e}+\cos s\,\mathbf{f}$ is a unit vector, because $\mathbf{e}$ and $\mathbf{f}$ are perpendicular unit vectors and $\sin^2 s+\cos^2 s=1$. Hence

$$AB=|\mathbf{a}-\mathbf{b}|=2d\,|\sin\tfrac{1}{2}(\alpha-\beta)|,$$

as required. (Geometrically: OAB is isosceles with equal sides d and apex angle $|\alpha-\beta|$; dropping the perpendicular from O gives the same answer.)

The angle with the vertical

When $\alpha\neq\beta$ we have $\sin\frac{1}{2}(\alpha-\beta)\neq 0$, since $0<|\alpha-\beta|<\frac{\pi}{2}$, so $A\neq B$ and the line AB is well defined. The calculation above shows that it is parallel to the unit vector $\mathbf{u}$, or equivalently to $\mathbf{v}=-\mathbf{u}=\sin s\,\mathbf{e}-\cos s\,\mathbf{f}$.

Ask yourself A line has two directions. Which one should we compare with the vertical?

The angle between a line and the vertical is the acute (or right) angle, so we use whichever direction along the line points upwards, or at least not downwards. Now

$$\mathbf{v}\cdot\mathbf{k}=\sin s\,(\mathbf{e}\cdot\mathbf{k})-\cos s\,(\mathbf{f}\cdot\mathbf{k})=\sin\theta\sin s.$$

Since $0\leq s<\frac{\pi}{2}$ and $0<\theta<\frac{\pi}{2}$, this is non-negative, so $\mathbf{v}$ is the upward direction. As $\mathbf{v}$ and $\mathbf{k}$ are both unit vectors, the angle ϕ between them satisfies

$$\cos\phi=\mathbf{v}\cdot\mathbf{k}=\sin\theta\sin\tfrac{1}{2}(\alpha+\beta),$$

which is the required result. Notice that it does not depend on d: the line AB moves parallel to itself as they walk.

Checks

- If Arthur walks horizontally ($\alpha = \frac{\pi}{2}$, the boundary case) and Bertha straight up ($\beta = 0$), then $AB = 2d \sin\frac{\pi}{4} = \sqrt{2}\,d$, which is Pythagoras for two perpendicular paths of length d. With $\theta = 30^\circ$ and $d = 1$, coordinates give $A = (1, 0, 0)$, $B = (0, \frac{\sqrt{3}}{2}, \frac{1}{2})$, so the vertical drop from B to A is $\frac{1}{2}$ over a distance $\sqrt{2}$: $\cos\phi = \frac{1}{2\sqrt{2}} \approx 0.354$, matching $\sin 30^\circ \sin 45^\circ$, so $\phi \approx 69.3^\circ$.
- If the plane is horizontal ($\theta = 0$), $\cos\phi = 0$: the line is horizontal, as it must be.
- A computer check with 10^5 random choices of θ, α, β, d in genuine 3D coordinates agreed with both formulae to within 10^{-12}.

Common slips

- Measuring the angle in the plane between AB and the steepest line instead of the angle in space with the vertical; θ must appear in the answer.
- Forgetting the modulus: $\sin\frac{1}{2}(\alpha - \beta)$ is negative when $\beta > \alpha$, but a distance is not.
- Taking the downward direction along AB and getting $\cos\phi = -\sin\theta \sin s$; the angle of a line with the vertical is never obtuse.

Beyond STEP

The method here is to choose an orthonormal basis adapted to the problem (two vectors in the plane, then relate them to the ambient vertical) and to read off lengths and angles from dot products. In a first-year linear algebra course this becomes the idea of changing basis and of orthogonal projection: $\mathbf{v} \cdot \mathbf{k}$ is the length of the projection of $\mathbf{v}$ onto the vertical. Strang's *Introduction to Linear Algebra* develops these ideas carefully, with plenty of geometry.

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